

Achieving prominent piezoelectric response and high Curie temperature in BiFeO₃-BaTiO₃-based ceramics through multi-component cooperative regulation and quenching treatment

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ARTICLE INFO

Keywords:

BiFeO₃-BaTiO₃

Piezoelectric property

High-temperature piezoelectric ceramics

Lead-free piezoelectric ceramics

Quenching treatment

ABSTRACT

The preparation of high-temperature lead-free piezoelectric ceramics with a large piezoelectric constant (d_{33}) and high Curie temperature (T_C) has long been the goal of materials researchers. In this paper, lead-free piezoelectric ceramics of $0.72\text{Bi}_{1.02}\text{FeO}_3\text{-}0.28\text{BaTiO}_3\text{-}0.015(\text{Ba}_{0.6}\text{Sr}_{0.4})\text{TiO}_3\text{-}x\text{Bi}(\text{Zn}_{0.5}\text{Ti}_{0.5})\text{O}_3$ (referred to as $0.72\text{BF-}0.28\text{BT-}0.015\text{BST-}x\text{BZT}$, $0.005 \leq x \leq 0.03$) were prepared by the solid-state method. We innovatively introduced $(\text{Ba}_{0.6}\text{Sr}_{0.4})\text{TiO}_3$ with a high dielectric constant into $0.72\text{BF-}0.28\text{BT}$ ceramics to enhance piezoelectric properties. Meanwhile, the high tetragonal degree $\text{Bi}(\text{Zn}_{0.5}\text{Ti}_{0.5})\text{O}_3$ is introduced to enhance lattice distortion, forming a local heterogeneity structure, which achieves the enhancement of piezoelectric response and thermal stability. And combined with the quenching treatment, the electrical properties of the ceramics were optimised. All the ceramic samples are located in the morphotropic phase boundaries (MPB) region where the rhombohedral (R) and pseudo-cubic (PC) phases coexist. The ceramic with optimal performance $0.72\text{BF-}0.28\text{BT-}0.015\text{BST-}0.015\text{BZT}$ ($d_{33} = 182 \text{ pC/N}$, $T_C = 529^\circ\text{C}$) exhibited enhanced piezoelectric activity ($d_{33} = 196 \text{ pC/N}$) and a higher Curie temperature ($T_C = 540^\circ\text{C}$) after quenching. This improvement can be attributed to the quenching treatment, which enhances lattice distortion and promotes the formation of long-range ordered electric domains. The new $0.72\text{BF-}0.28\text{BT-}0.015\text{BST-}x\text{BZT}$ ceramics designed in this study provide a new idea for the preparation of high-performance lead-free piezoelectric materials.

1. Introduction

Piezoelectric ceramics are a type of functional material that can realise the mutual conversion of electrical energy and mechanical energy, and they play a significant role in electronic devices such as resonators, ultrasonic transducers, and energy harvesters [1,2]. Currently, $\text{Pb}(\text{Zr}, \text{Ti})\text{O}_3$ (PZT) piezoelectric ceramics are widely utilised in various applications due to their superior electromechanical performance and excellent thermal stability, maintaining a dominant position in the

commercial piezoelectric materials market. However, the toxic lead content in PZT exceeds 60 wt%, and even in $0.64\text{PT-}0.36\text{BS}$, it reaches as high as 43.82 wt% [3]. As an electronic product, it will cause serious hazards to human health and the ecological environment due to the high content of harmful substances in the process of large-scale use or recycling. Therefore, the more environmentally friendly lead-free piezoelectric ceramics have attracted people's attention, and related reports have been increasing [4–8].

Among them, the BiFeO_3 -based ceramics have captured the interest

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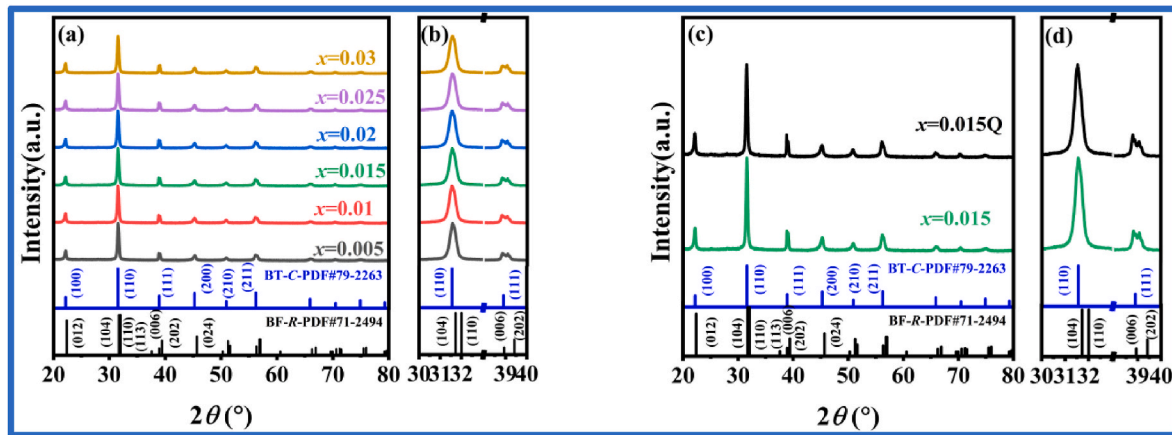


Fig. 1. The XRD diffraction pattern of 0.72BF-0.28BT-0.015BST-xBZT ceramics (a) $0.005 \leq x \leq 0.03$ samples, (b) 30° - 40° Localised magnified diffraction pattern, (c) $x = 0.015$, $x = 0.015Q$ samples, and (d) 30° - 40° Localised magnified diffraction pattern.

of numerous researchers due to their high Curie temperature of up to 830°C and large spontaneous polarisation ($P_r = 90$ - $100\ \mu\text{C}/\text{cm}^2$) [9,10]. This is mainly attributed to the hybridisation of $\text{Bi}^{3+} 6s^2$ and $\text{O}^{2-} 2p$ orbitals, which results in significant polarisation and strain [11]. However, pure BF ceramics are difficult to synthesise by conventional solid-state reaction and are prone to generating impure phases. Besides, the volatilisation of Bi, the valence change of Fe ions ($\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$) and the formation of oxygen vacancy ($\text{V}_\text{O}^\bullet$) during the sintering process, which result in high leakage conductivity and low sample performance in BF ceramics [12,13]. For address these issues, the researchers chose the ABO_3 -type perovskite BaTiO_3 , which has a high dielectric constant and excellent insulation properties, to form a solid solution with it. The constructed BiFeO_3 - BaTiO_3 (BF-BT) ceramics have high dielectric properties and high Curie temperatures.

To further improve the performance of BiFeO_3 - BaTiO_3 (BF-BT) ceramics, researchers have achieved significant enhancement in electrical properties by constructing morphotropic phase boundaries (MPB) where multiple phases coexist. Guan et al. [14]. studied $(1-x)\text{BiFeO}_3$ - $x\text{BaTiO}_3$ (BF-xBT) ceramics and constructed MPB through fine coordination of phase boundaries. It was found that the 0.68BF-0.32BT ceramics exhibited good ferroelectric properties and thermal stability: $d_{33} = 185\ \text{pC}/\text{N}$, $P_r = 33.152\ \mu\text{C}/\text{cm}^2$, $T_C = 470^\circ\text{C}$. Additionally, Zhou et al. [15]. synthesising the $0.70\text{Bi}_{(1+x)}\text{Fe}_{0.97}\text{Al}_{0.03}\text{O}_3$ - 0.30BaTiO_3 ($0 \leq x \leq 0.05$) ceramics, the piezoelectric properties and insulation of ceramics were improved by adjusting bismuth compensation and polarisation process, which obtained $\text{Bi}_{1.02}\text{FA}$ -BT ceramics with excellent electromechanical properties ($d_{33} = 196\ \text{pC}/\text{N}$, $k_p = 34\%$, $Q_m = 32$) and a high Curie temperature ($T_C = 480^\circ\text{C}$), it provides a new research idea to solve the problem of high conductivity caused by various defects.

At the same time, researchers adjust the local composition and structure by adding relaxation end elements, introducing PNRs to enhance the electromechanical conversion capability of materials; design and utilise the synergistic effect of third and fourth group elements to construct new multi-component solid solution systems [16]. For example, Luo et al. [17]. constructed the BF-BT-BST ternary system ceramics by introducing $(\text{Ba}_{0.6}\text{Sr}_{0.4})\text{TiO}_3$ (BST) with composition disorder and diffuse phase transition characteristics, which breaks the typical long-range ferroelectric domain-ordered structure in ferroelectric ceramics, promote the formation of PNRs, and comprehensively improves the electrical properties of materials. Yang et al. [18]. successfully prepared 0.7BF-0.3BT-xBZT lead-free piezoelectric ceramics by introducing the superquadratic material of $\text{Bi}(\text{Zn}_{0.5}\text{Ti}_{0.5})\text{O}_3$ (BZT) as the third component. Thought controlling the phase composition, enhancing lattice distortion and forming local nano-heterogeneous structures, realises the simultaneous enhancement of piezoelectric internal and external contributions, which significantly improves the

high-temperature piezoelectric activity and thermal stability of ceramics. Meanwhile, the researchers designed multicomponent $\text{BF}31\text{BT}$ -0.005(BNT -20BKT)-xBZT ceramics, forming nanodomains and abundant polar nanoregions (PNRs), which achieved synchronous enhancement of ultra-high temperature stability and ultra-high piezoelectric response [16]. Furthermore, some studies have shown that heat treatment is an effective method for enhancing the electrical properties of piezoelectric ceramics [19]. Chang et al. [20]. prepared BF-B (BN)_{0.015}T ceramics with both high Piezoelectric property ($d_{33} = 196\ \text{pC}/\text{N}$) and high Curie temperature ($T_C = 498^\circ\text{C}$) by controlling ceramic lattice distortion and magnetic domain structure through quenching processes. Concurrently, Quenching treatment can also be used to control the phase composition of ceramics. Zheng et al. [21]. demonstrated in their study of $\text{BF}(\text{ZS})_{0.003}\text{-B}(\text{BN})_x\text{T}$ ceramics that quenched samples exhibited a significantly increased content of the rhombic-phase (R-phase) alongside heightened lattice distortion, which enables the domain structure enhancement, thus effectively improving the electrical properties of the material.

In this study, a novel quaternary lead-free piezoelectric ceramic system was designed with the compositional formula $0.72\text{Bi}_{1.02}\text{FeO}_3$ - 0.28BaTiO_3 - $0.015(\text{Ba}_{0.6}\text{Sr}_{0.4})\text{TiO}_3$ - $x\text{Bi}(\text{Zn}_{0.5}\text{Ti}_{0.5})\text{O}_3$ ($0.005 \leq x \leq 0.03$). By introducing $(\text{Ba}_{0.6}\text{Sr}_{0.4})\text{TiO}_3$ (BST), which has a high dielectric constant, to cause compositional fluctuations and local heterogeneity, break the long-range order, form polar nanoregions, and thereby enhance the piezoelectric activity [22]. Meanwhile, introducing the high tetragonal degree and high spontaneous polarisation ($P_s = 103\ \mu\text{C}/\text{cm}^2$) $\text{Bi}(\text{Zn}_{0.5}\text{Ti}_{0.5})\text{O}_3$ (BZT) solid solution ($c/a = 1.21$) to adjust phase ratios, increase oxygen octahedral distortion and form nano-domain structures, thereby enhancing the material's overall performance [18,23]. In addition, the samples with the optimal performance were quenched to enhance the long-range ferroelectric order, resulting in higher piezoelectric activity ($d_{33} = 196\ \text{pC}/\text{N}$) and higher working temperature ($T_C = 540^\circ\text{C}$). This research provides a new strategy for the preparation of lead-free piezoelectric ceramics with excellent electrical properties and high Curie temperature.

2. Experimental procedures

2.1. Sample preparation

$0.72\text{Bi}_{1.02}\text{FeO}_3$ - 0.28BaTiO_3 - $0.015(\text{Ba}_{0.6}\text{Sr}_{0.4})\text{TiO}_3$ - $x\text{Bi}(\text{Zn}_{0.5}\text{Ti}_{0.5})\text{O}_3$ (referred to as 0.72BF-0.28BT-0.015BST-xBZT, $0.005 \leq x \leq 0.03$) ceramics were prepared by the traditional solid-state reaction method using Bi_2O_3 , Fe_2O_3 , BaCO_3 , ZnO , SrCO_3 and TiO_2 raw materials with a purity of 99.9 %. The powders were obtained by grinding the weighed powder and zirconia ball in anhydrous ethanol for 24h and drying in an

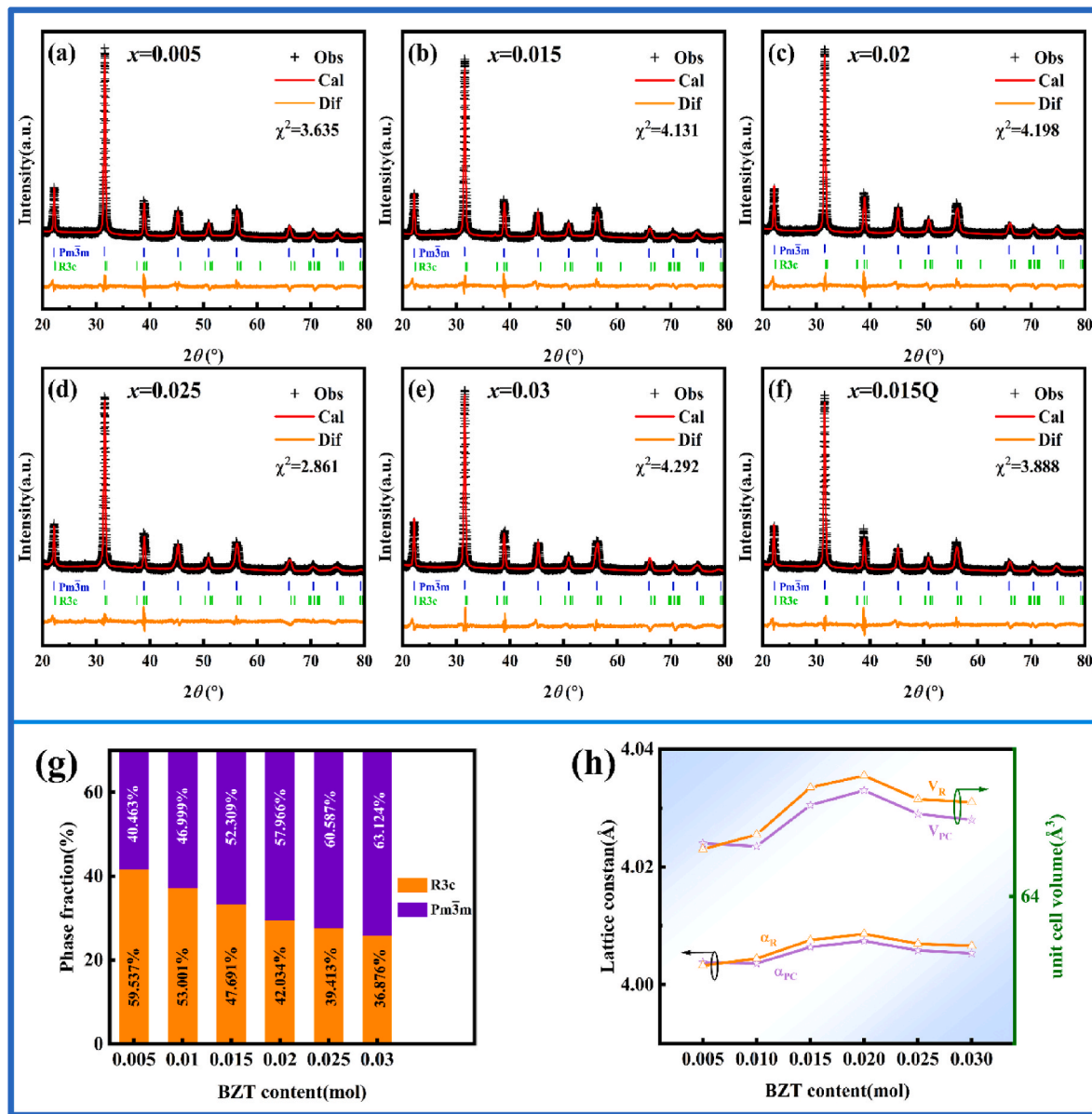


Fig. 2. Rietveld refinement patterns of 0.72BF-0.28BT-0.015BST-*x*BZT ceramics (a) *x* = 0.005, (b) *x* = 0.015, (c) *x* = 0.02, (d) *x* = 0.025, (e) *x* = 0.03, (f) *x* = 0.015Q. (g) The variation of the phase proportion for 0.72BF-0.28BT-0.015BST-*x*BZT (0.005 ≤ *x* ≤ 0.03) ceramics with the doping amount *x*; (h) lattice parameters and cell volumes for phase structure for 0.72BF-0.28BT-0.015BST-*x*BZT (0.005 ≤ *x* ≤ 0.03) ceramics.

oven at 95 °C. The powder was calcined at 800 °C for 4h at a heating rate of 2 °C/min. And 8 wt% polyvinyl alcohol (PVA) was added to the calcined powder for granulation, and the prepared powder was pressed into a disc with a diameter of 12 mm under a uniaxial pressure of 13 MPa. Put the pressed disc sample into the tubular furnace for sintering. Sintered at 600 °C for 3h in air, removed the binder, and sintered at 980 °C for 6h. The sintered samples were coated with electrodes on both sides and cured at 600 °C for 30 min. Finally, the ceramic samples were polarised in silicone oil at 110 °C for 15-20 min under a 5 kV/cm DC electric field. After standing for 24h, the samples with the best performance were quenched. The specific operation is to directly air quench the sample after holding it in a 980 °C tubular furnace for 6h (the quenched sample is expressed as 0.015Q).

2.2. Sample characterisation and testing

The phase structure of ceramics was analysed by X-ray diffractometer (D8 advance, Bruker Inc., Karlsruhe, Germany), and the XRD

diffraction data were refined by GASA software. The internal molecular vibrations and crystal structure of the materials were characterised and analysed using a Raman spectrometer (HR800 Evolution, JY, French). The microstructure of 0.72BF-0.28BT-0.015BST-*x*BZT ceramics was observed by Scanning Electron Microscopy (FEG-450, FEI, USA). Quasi-static *d*₃₃ tester (ZJ-3A, Institute of Acoustics, Chinese Academy of Science, China) was used to test the piezoelectric properties of samples at room temperature. At the same time, In-situ depolarisation equipment (wide temperature range *d*₃₃ instrument, TZFD-900, Harbin Julang Technology Co., Ltd, Harbin, China) is used to measure the piezoelectric properties of ceramics at high temperature in real time. The dielectric property temperature curve was measured using an LCR analyser (Keysight, 4980A). The hysteresis loop and current curve were measured by the ferroelectric measurement instrument (Huace FE-2000, Beijing Huace Test Instrument Co., Ltd, Beijing, China). The domain structure of the material was observed using a Transmission Electron Microscope (TEM, JEOL-2200 fs, JEOL, Japan).

Table 1

Rietveld refinement results of 0.72BF-0.28BT-0.015BST-xBZT ($0.005 \leq x \leq 0.03$) ceramics for phase structure, lattice parameters, unit cell volume and reliability factor.

Composition	Phase fraction(%)	Lattice Parameters a/c (Å)	Volume (V_R/V_{PC}) (Å ³)	Fitting Parameter R_{wp}/R_p
$x = 0.005$	R3c 59.537	5.651254	384.426	$R_{wp} = 0.0519$
	Pm $\bar{3}$ m 40.463	13.899302	65.025	$R_p = 0.0374$
		4.021235		$\chi^2 = 3.635$
$x = 0.01$	R3c 53.001	5.634870	381.850	$R_{wp} = 0.0484$
	Pm $\bar{3}$ m 46.999	13.886560	65.223	$R_p = 0.0349$
		4.0235329		$\chi^2 = 3.169$
$x = 0.015$	R3c 47.691	5.642508	383.652	$R_{wp} = 0.0557$
	Pm $\bar{3}$ m 52.309	13.914337	65.049	$R_p = 0.0405$
		4.021742		$\chi^2 = 4.131$
$x = 0.02$	R3c 42.034	5.640677	382.992	$R_{wp} = 0.0566$
	Pm $\bar{3}$ m 57.966	13.899411	64.923	$R_p = 0.0404$
		4.019146		$\chi^2 = 4.198$
$x = 0.025$	R3c 39.413	5.638870	382.580	$R_{wp} = 0.0464$
	Pm $\bar{3}$ m 60.587	13.893270	65.105	$R_p = 0.0341$
		4.022892		$\chi^2 = 2.861$
$x = 0.03$	R3c 36.876	5.64244	382.420	$R_{wp} = 0.0572$
	Pm $\bar{3}$ m 63.124	13.87001	64.764	$R_p = 0.0416$
		4.015846		$\chi^2 = 4.292$
$x = 0.015Q$	R3c 55.121	5.647165	384.134	$R_{wp} = 0.0550$
	Pm $\bar{3}$ m 44.879	13.908864	64.965	$R_p = 0.0401$
		4.020004		$\chi^2 = 3.888$

3. Results and discussion

The piezoelectric properties of BF-BT ceramics are closely correlated with their crystal structure evolution. Fig. 1(a) shows the XRD patterns of 0.72BF-0.28BT-0.015BST-xBZT ($0.005 \leq x \leq 0.03$) ceramics; all the ceramic samples showed a standard perovskite structure and no impurity phase, indicating that BST and BZT were completely dissolved in the main phase. Fig. 1(b) presents the local magnified X-ray diffraction patterns at 30° - 33° and 38° - 40° . With the introduction of $(ZT)^{3+}$ ions, there is a diffraction peak near 31° , and the diffraction peak near 39° shows continuous splitting, indicating that all the ceramics have a morphotropic phase boundary (MPB) structure where the R-phase and the PC-phase coexist. Fig. 1(c) shows the X-ray diffraction patterns of the sample $x = 0.015$ and the quenched sample $x = 0.015Q$. The results show that the quenched sample ($x = 0.015Q$) still maintains the perovskite structure. Fig. 1(d) provides the local amplified X-ray diffraction patterns of two samples. Similar to the unquenched sample, the quenched sample ($x = 0.015Q$) also shows the coexistence of R phase and PC phase. To further study the crystal structure changes of ceramics, the GSAS Rietveld method was employed to refine the X-ray diffraction data of ceramics. Fig. 2(a)–(f) shows the Rietveld refinement results; the R3c and Pm $\bar{3}$ m two-phase models produce relative reliability factors. As shown in Fig. 2(g), with the increase of x , the proportion of the PC phase increases from 40.463 % to 63.124 %. From the Rietveld refinement results, it can be seen that the ratio of R phase to PC phase in the sample with $x = 0.015$ is 48/52. The sample ($x = 0.015Q$) after quenching treatment, the ratio of R phase to PC phase is 55/45, which indicates quenching promotes the phase transformation from PC phase to R phase. Similar experimental results were reported in previous studies [21,24]. Fig. 2(h) shows that the changes in lattice parameters and cell volume of the R phase and PC phase of the test sample are the same, and the addition of a small amount of BZT will lead to slight lattice expansion. This might be caused by the fact that the ionic radius of the introduced Zn^{2+} (0.74 Å, CN = 6) is larger than that of the Fe^{3+} (0.64 Å, CN = 6) at the B site in the crystal lattice, resulting in a slight lattice expansion. Table 1 summarises the detailed structural parameters derived from a Rietveld refinement of the ceramic samples.

To gain a deeper insight into the local microstructure of 0.72BF-0.28BT-0.015BST-xBZT ($0.005 \leq x \leq 0.03$) ceramics, Raman spectroscopy testing was performed on the ceramic materials. The BiFeO₃ material with R-phase structure theoretically has 13 Raman vibration modes $\Gamma_{Raman} = 4A_1 + 9E(R3c, C_{3v}^6, Z = 2)$ [25,26]. However, phase

transitions occur within the ceramics, causing some of the vibrational modes to overlap or disappear [16]. Fig. 3(a) shows the room temperature Raman spectra of 0.72BF-0.28BT-0.015BST-xBZT ($0.005 \leq x \leq 0.03$) ceramics, which are mainly divided into two regions: (1) $150 \text{ cm}^{-1} \leq \lambda \leq 400 \text{ cm}^{-1}$, The Raman vibration mode in this region is mainly derived from the vibration of the B-site cation, which is specifically manifested as the telescopic vibration of the B-O bond. (2) In the high wavenumber region of $\lambda > 400 \text{ cm}^{-1}$, its vibration mode reflects the structural characteristics of the BO₆ octahedron, especially its tilt degree. In the low wavenumber region of $\lambda \leq 150 \text{ cm}^{-1}$ in ceramics, the pure BiFeO₃ material has an obvious frequency shift due to the hybridization of bismuth and oxygen, while there is no obvious shift in 0.72BF-0.28BT-0.015BST-xBZT ceramics, which may be due to the introduction of Ba²⁺ and Sr²⁺ to break the Bi-O orbital hybridization, resulting in the weakening of the interaction between A-site cation and O²⁻. As shown in Fig. 3(b), in the wavenumber range of $150 \text{ cm}^{-1} \leq \lambda \leq 400 \text{ cm}^{-1}$, the A₁₋₂ vibrational mode exhibits a red shift with increasing x , which is attributed to the weakening of the hybridisation between the p orbitals of O²⁻ and the d orbitals of Fe³⁺, Ti⁴⁺, and Zn²⁺. Such a suppression of orbital hybridisation promotes the formation of nano-scale polar regions, disrupts the long-range ferroelectric order, and thereby induces a macroscopic transformation of the ceramics from a normal ferroelectric to a relaxor ferroelectric [27]. For the high wavenumber region ($\lambda > 400 \text{ cm}^{-1}$), the distortion of the BO₆ octahedron (internal tilting of the BO₆) is closely related to the Raman vibration modes of A₁₋₄ and E-7, which can be shown by the Raman intensity ratio $I(E-7)/I(A_1-4)$. Fig. 3(c) shows the Raman intensity ratio decreases as a whole with the increase of x value, indicating that the distortion degree of the BO₆ octahedron decreases [28], which for the perspective of local structure evolution, the radii of the doped ions Ba²⁺ (1.35 Å, CN = 6), Sr²⁺ (1.18 Å, CN = 6), and Zn²⁺ (0.74 Å, CN = 6) are larger than the ionic radii of the A-site Bi³⁺ (1.03 Å, CN = 6) and B-site Fe³⁺ (0.64 Å, CN = 6) in the original system [16,27]. This co-doping at the A-site and B-site may cause the crystal cell expansion of BF-BT-based ceramics, reducing the degree of structural distortion of the BO₆ oxygen octahedron, thereby leading to the instability of the crystal structure and promoting the refinement of the electric domain size. Fig. 3(d) shows the B-O bond vibration model and the internal distortion diagram of the BO₆ octahedron.

In BF-BT ceramics, an appropriate grain size can facilitate the movement of domain walls and reduce internal stress, thereby effectively enhancing the piezoelectric properties. Fig. 4(a)–(f) presents the

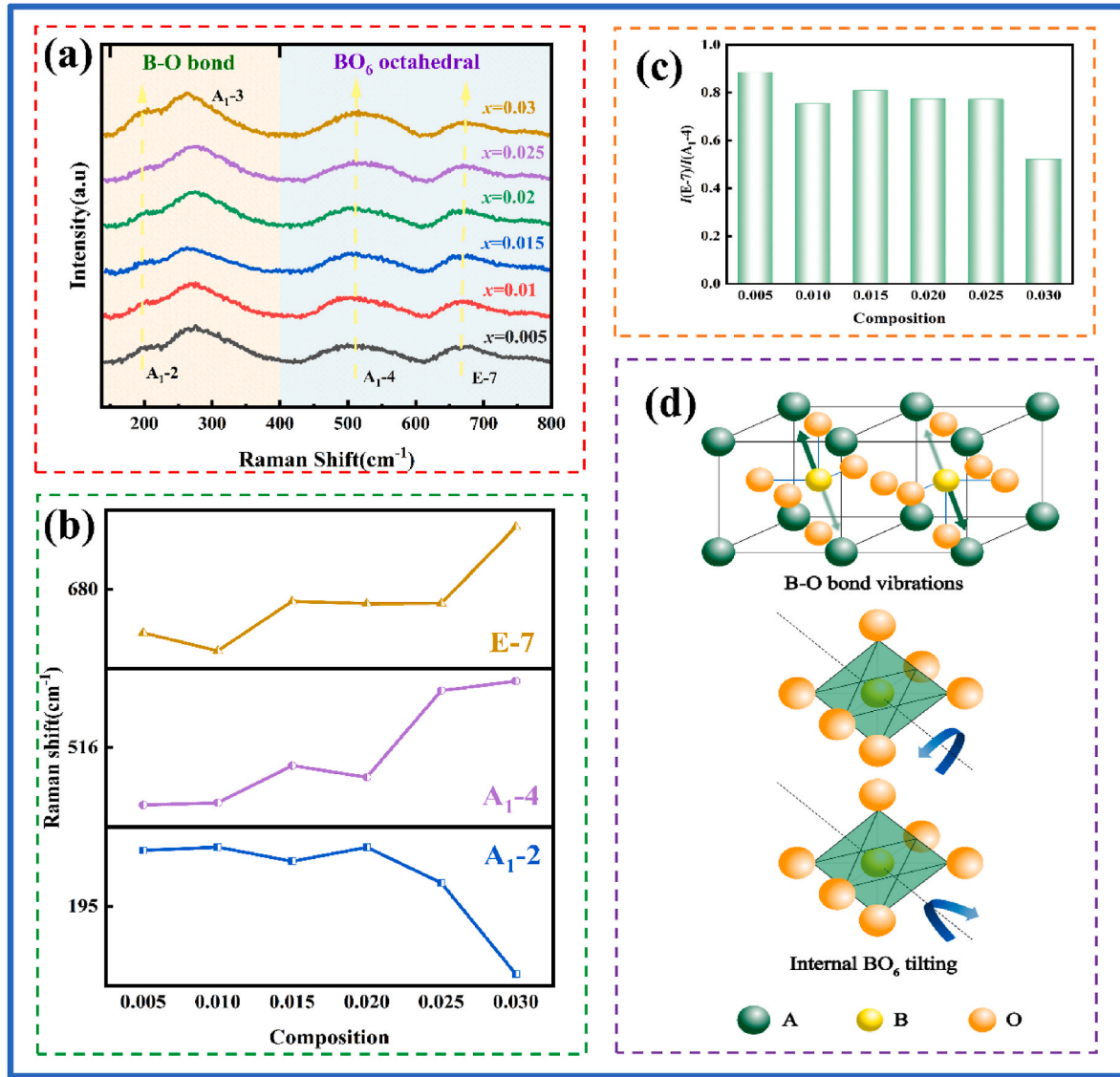


Fig. 3. (a) Raman Spectra of 0.72BF-0.28BT-0.015BST- x BZT ($0.005 \leq x \leq 0.03$) ceramics at Room Temperature; (b) the Raman shift of individual mode peaks; (c) the intensity ratio of $I(E-7)/I(A_1-4)$; (d) the B-O bond vibration model diagram and BO₆ internal distortion diagram.

microstructure and grain size diagrams of the 0.72BF-0.28BT-0.015BST- x BZT ($0.005 \leq x \leq 0.03$) ceramic section. All samples exhibited a dense microstructure with well-defined grain boundaries and no significant holes. As x increases, the average grain size of the sample shows an upward trend, with the maximum average grain size obtained being 5.9 μm at $x = 0.03$. The average grain size of ceramics is closely related to grain boundary migration behaviour. A small amount of ZnO and Bi₂O₃ as sintering additives helps to reduce the sintering temperature and promote the formation of a liquid phase, to accelerate the migration of grain boundaries, and finally realise the densification of sintering and the increase of grain size [16]. For perovskite ferroelectrics, when the grain size undergoes a simple evolution, the size of the electric domain also exhibits corresponding evolution behaviour. The size of the electric domain (d) is constrained by the grain size (g), and there exists a certain functional relationship as follows:

$$d = \left\{ \left(\frac{\sigma}{\epsilon^* P_s^2 g} \right) \right\}^{1/2} \quad (1)$$

Where P_s , ϵ^* and σ represent the spontaneous polarisation, effective dielectric constant and energy density of the domain wall, respectively. The larger grain size provides more free space for the formation and

growth of domains while reducing the pinning effect of grain boundaries on domain wall motion, thereby promoting domain wall migration [29]. Fig. 4(g)–(i) shows the EDS images of 0.72BF-0.28BT-0.015BST-0.015BZT ceramics. The Sr and Zn elements are uniformly distributed throughout the ceramics, indicating that BST and BZT have successfully combined with the BF-BT base ceramics, confirming the formation of effective solid solutions.

The piezoelectric properties of BF-BT ceramics are closely related to the dielectric temperature spectra, which can reflect the phase transition temperature and polar relaxation behaviour of the ceramics, thereby revealing the intrinsic origin and temperature stability of the piezoelectric properties. Fig. 5(a)–(f) provides the temperature variation spectra of dielectric constant and dielectric loss of 0.72BF-0.28BT-0.015BST- x BZT ($x = 0.005, 0.015, 0.02, 0.025, 0.03, 0.015Q$) ceramics at different frequencies. Within the temperature range from room temperature to 350 $^{\circ}\text{C}$, all the ceramic samples exhibited normal ferroelectric properties. The dielectric constant changed gradually with the increase in temperature, and the dispersion phenomenon at different frequencies was weak, indicating that the ceramic materials maintained good long-range order within this temperature range and possessed excellent dielectric stability. When the temperature rises to around the

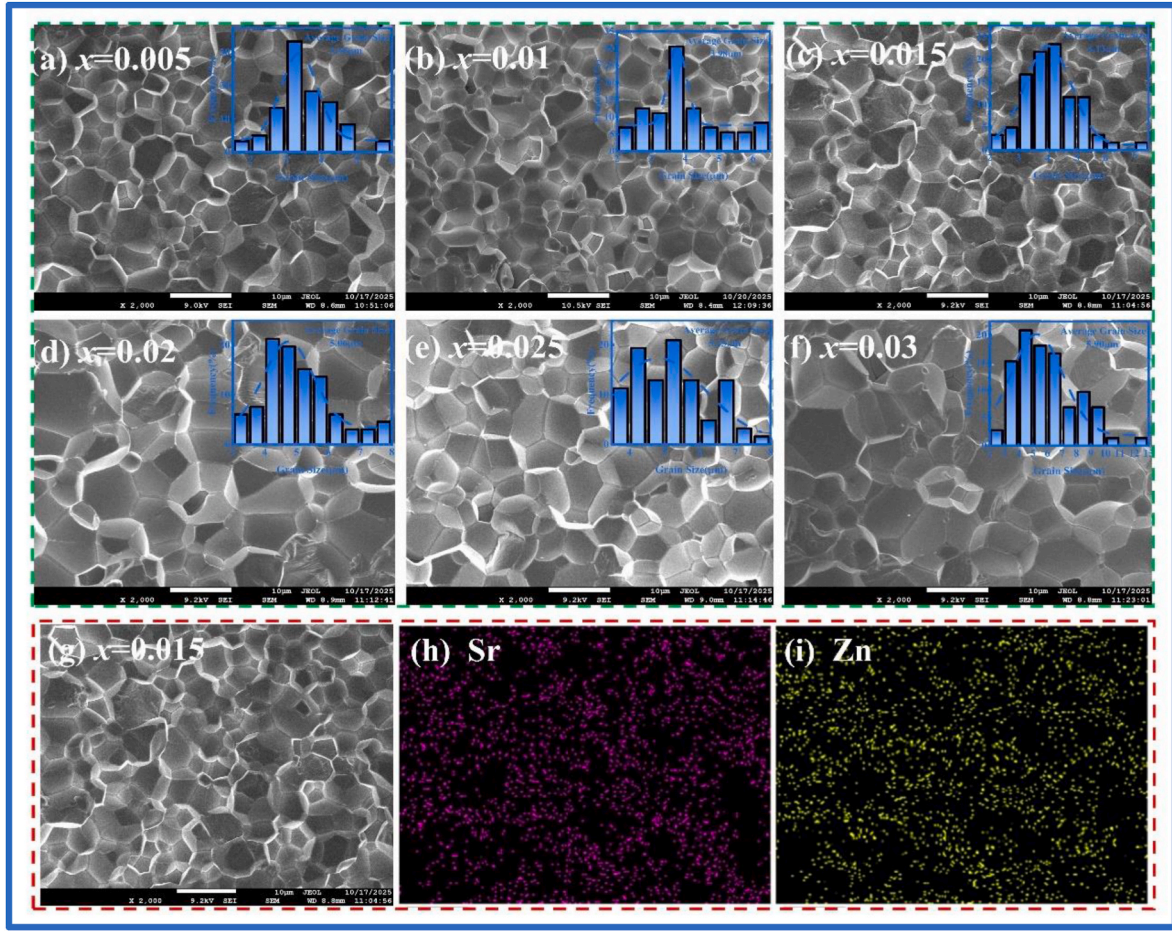


Fig. 4. (a)–(f) SEM images of the 0.72BF-0.28BT-0.015BST- x BZT ($0.005 \leq x \leq 0.03$) ceramic cross-section. 0.72BF-0.28BT-0.015BST-0.015BZT ceramic: (g) microscopic morphology of the cross-section; (h) Distribution map of Sr element; (i) Distribution map of Zn element.

Curie temperature (T_C), a sharp dielectric anomaly peak can be observed. In the 0.72BF-0.28BT-0.015BST- x BZT ceramic, Ba, Sr, and Bi simultaneously occupy the A positions, while Ti, Fe, and Zn ions jointly occupy the B positions, which by employing the strategy of heterovalent ion co-doping at the A and B positions, a lattice chemical-scale inhomogeneity was successfully constructed in the ceramic, promoting the formation of short-range ordered microregions and generating the dielectric anomaly peak. The T_C of all ceramic samples exceeds 500 °C, indicating that the dielectric and ferroelectric properties of the materials have high temperature stability. The temperature (T_m) corresponding to the maximum dielectric constant increases first and then decreases with the increase of x , which may be due to the competitive effect between the reduction of R phase content and its lattice distortion [16]. Meanwhile, the lattice distortion caused by quenching treatment increases, which further increases the Curie temperature (T_C) of the sample to 540 °C, and the dielectric peak becomes sharper. The trend of sample dielectric loss with temperature is similar to that of dielectric constant. The dielectric constant (ϵ_r) and dielectric loss ($\tan\delta$) of all samples decrease with the increase of frequency. Above 400 °C, $\tan\delta$ increases rapidly, which is mainly due to the enhanced leakage conductance behaviour at high temperatures. To further understand the ceramic dielectric relaxation phenomenon, the dispersion coefficient (γ) can be calculated using the modified Curie-Weiss law [30]:

$$\frac{1}{\epsilon} - \frac{1}{\epsilon_m} = \frac{(T - T_m)^\gamma}{C} \quad (2)$$

Among them, ϵ_m represents the maximum dielectric constant, T_m is the phase transition temperature, and C is the Curie-Weiss constant.

When the $\gamma = 1$, the ceramics are normal ferroelectric; when the $\gamma = 2$, they are relaxor ferroelectric [31,32]. The γ values of all samples are between 1 and 2, which proves that the samples are relaxor ferroelectrics. After quenching, the γ value decreased from 1.68 ($x = 0.015$) to 1.5828 ($x = 0.015Q$), which reduction in γ value indicates that the quenching treatment has reduced the ceramic relaxation properties, and the long-range ferroelectric order of the material has been enhanced, promoting it to transform from the relaxation state to the normal ferroelectric [21].

Many literatures indicate that the high leakage current and high dielectric loss of BF-BT-based ceramics are mainly attributed to the change in the valence state of iron ions and the accompanying generation of charged defects such as oxygen vacancies ($V_O^\bullet$) [33,34]. To deeply investigate the defect structure and evolution pattern of the 0.72BF-0.28BT-0.015BST- x BZT ceramic sample, we conducted X-ray photoelectron spectroscopy (XPS) measurements. Fig. 6(a)–(c) displays the XPS spectra of ceramic Fe 2p. All test samples have two wide and asymmetric peaks, which confirms the coexistence of Fe^{2+} and Fe^{3+} . This phenomenon mainly results from the restoration of some Fe^{3+} to Fe^{2+} during the high-temperature sintering process, which valence change behaviour maintains the local charge balance of the system by forming oxygen vacancies ($V_O^\bullet$), which is a typical charge compensation mechanism in BF-based ceramics. At Fe2p_{1/2}, the binding energies of Fe^{2+} and Fe^{3+} are about 709 eV and 711 eV, respectively. The content of Fe^{2+} in the test sample decreased from 31.14% ($x = 0.005$) to 24.27% ($x = 0.015$), and for the quenched sample ($x = 0.015Q$), the content of Fe^{2+} further dropped to 15.84%, which indicates that the quenching treatment has an inhibitory effect on the valence change of iron ions. The XPS

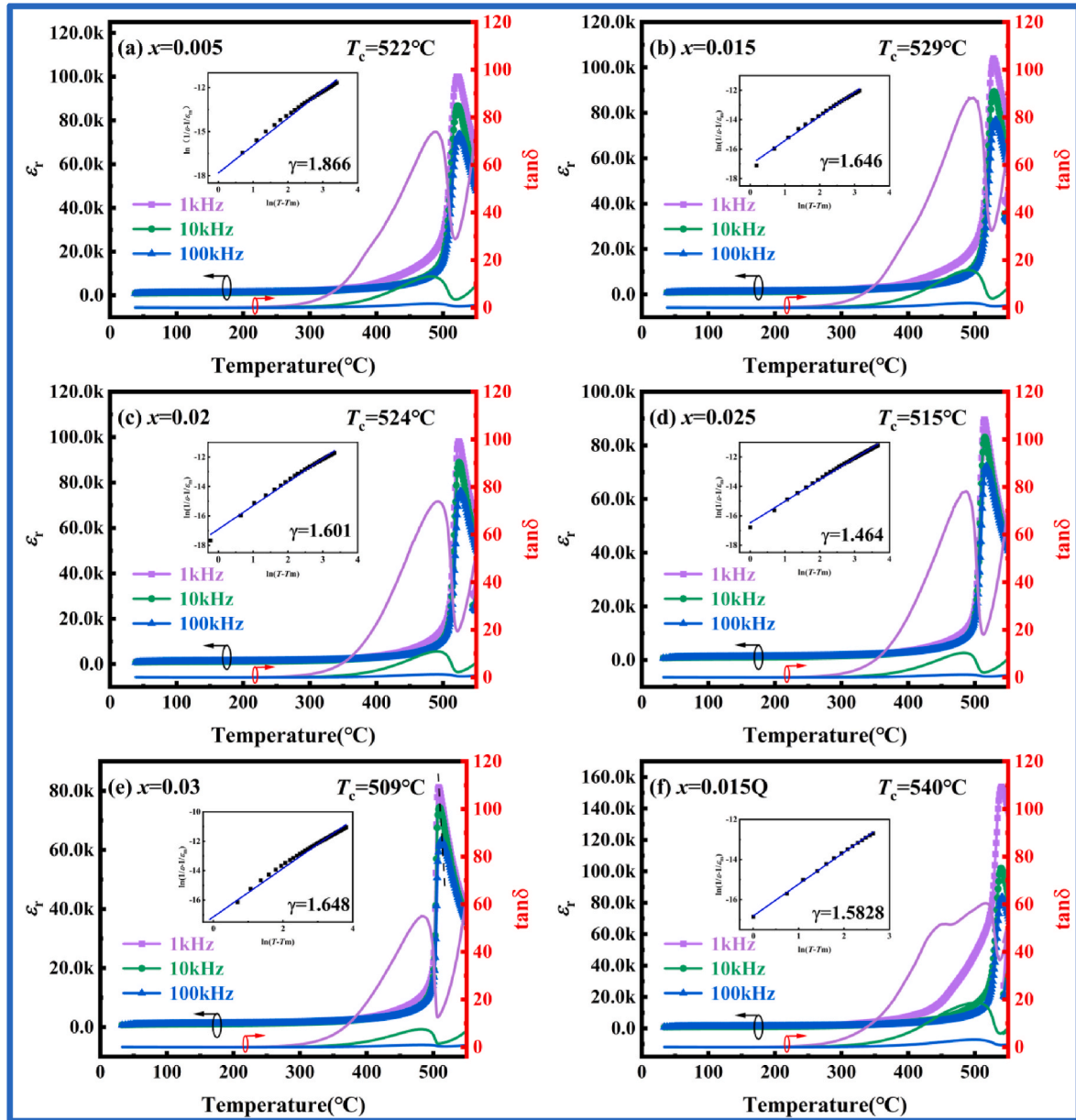


Fig. 5. (a)–(f) 0.72BF-0.28BT-0.015BST- x BZT ($x = 0.005, 0.015, 0.02, 0.025, 0.03, 0.015Q$) ceramics dielectric temperature spectrum and dispersion coefficient diagram drawn.

spectra of ceramics O 1s spectra are shown in Fig. 6(d)–(f), in which the O_L peak (~ 529 eV) with the lowest binding energy is attributed to the lattice oxygen in the perovskite lattice, the O_v peak (~ 531 eV) with the middle binding energy derived from the existence of oxygen vacancies, and the O_{O-C} peak (~ 532 eV) with the highest binding energy is related to the C-O bond. As x increases, the concentration of oxygen vacancies decreases from 38.09% ($x = 0.005$) to 23.93% ($x = 0.015$). A small amount of ZnO as a sintering aid reduces the sintering temperature, thereby reducing the defect concentration. In addition, after quenching treatment, the concentration of oxygen vacancies further drops to 19.41%, which may be because the quenching treatment retains the low oxygen vacancy state of the ceramic at high temperatures and prevents the migration and aggregation of defects during the cooling process; thus, the sample exhibits fewer oxygen vacancy defects at room temperature.

Ferroelectric measurement is a key method to characterise the ferroelectric properties, domain structure, and polarisation behaviour of BF-BT ceramics, reflecting the intrinsic relationship between

ferroelectric characteristics and piezoelectric performance. Fig. 7(a) and (b) show the hysteresis loop (P - E) and current loop (I - E) of 0.72BF-0.28BT-0.015BST- x BZT ($0.005 \leq x \leq 0.03, x = 0.015Q$) ceramics measured at room temperature. The results show that the hysteresis loops (P - E) of all test samples show a saturated shape under the electric field of 50 kV/cm, which has typical ferroelectric characteristics. With the increase of x , the P_r increases first and then decreases. This might be because the increase in BZT content leads to lattice distortion, which improves the ferroelectric activity of the material. However, the introduction of mixed valence cationic at positions A and B increases the disorder within the ceramic material, which breaks the original long-range ferroelectric order, resulting in the reduction of P_r [16]. At the same time, it is also observed that the samples treated by quenching ($x = 0.015Q$) show higher residual polarisation and lower coercive field (E_c), because the quenching treatment inhibits the enrichment of defects (such as $V_{O}^{\bullet\bullet}$) formed at high temperature to the domain wall during cooling, thus reducing the pinning effect on the domain wall, which makes the domain easier to flip under the external electric field, so the

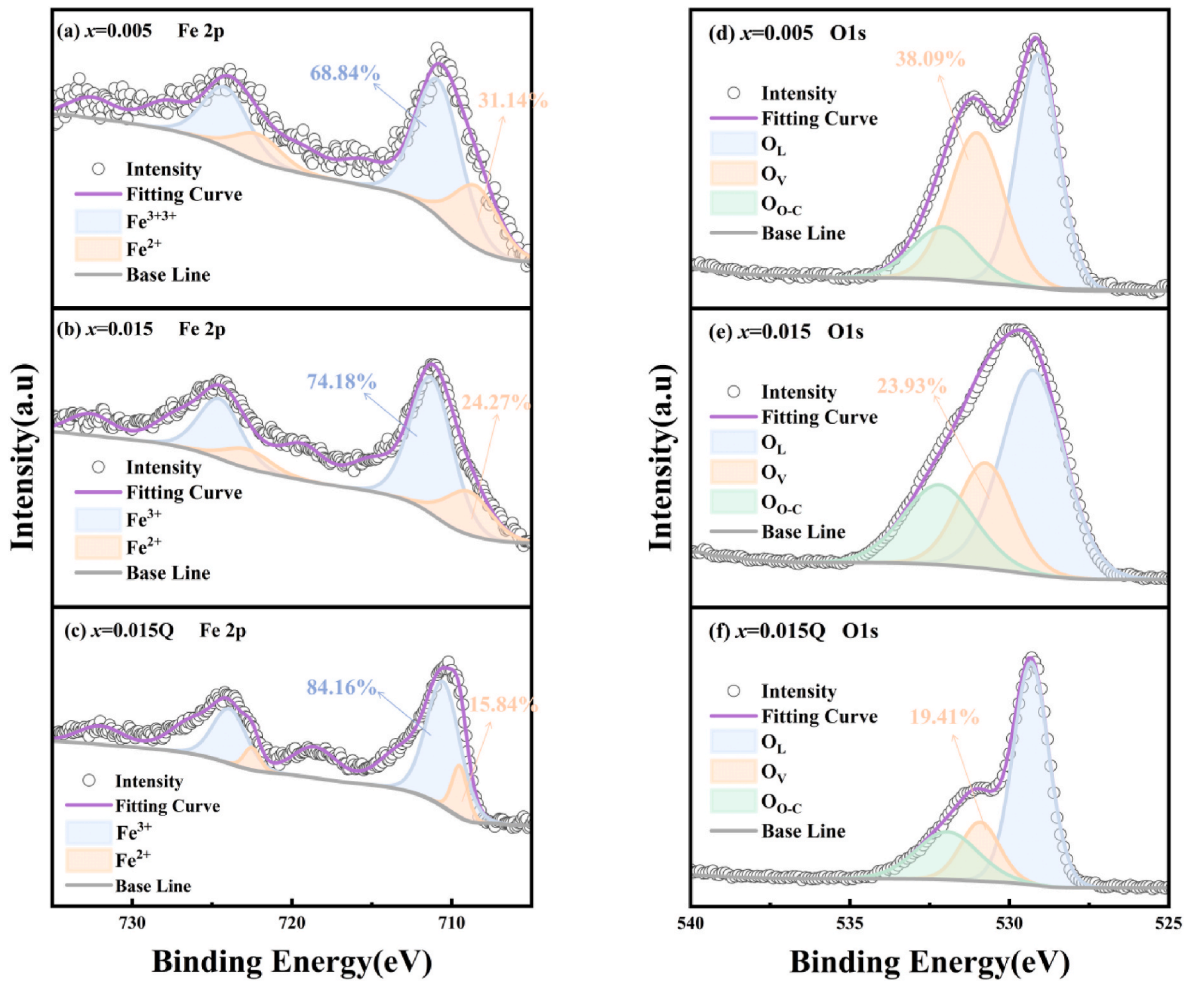


Fig. 6. (a)–(f) X-ray photoelectron spectra of Fe 2p and O 1s for 0.72BF-0.28BT-0.015BST-xBZT ($x = 0.005$, $x = 0.015$, $x = 0.015Q$) ceramic.

E_C is reduced [35]. Moreover, quenching also induces the transformation of crystal structure from PC phase to R phase, which promotes the formation of long-range ordered macro ferroelectric domains, leading to the increase of P_r in quenched samples [36]. In Fig. 7(b), all the test samples exhibited the standard double-peak I - E curve. The sharp peak current in the I - E curve is caused by the magnetic domain transformation in the coercive field, reflecting the degree of electric domain flipping. It can be observed that the quenched samples have the highest current peak, further indicating that the domain wall pinning effect in the quenched samples is the weakest, and the electric domain flipping dynamics are more active.

Fig. 7(c)–(f) shows the variable temperature P - E loop and I - E curve of samples $x = 0.015$ and $x = 0.015Q$ under the test conditions of 50 kV/cm electric field, 1Hz frequency and different temperature ranges (20 °C–110 °C). Fig. 7(g)–(i) presents the temperature-dependent diagrams of P_r , P_{max} , and E_C for the $x = 0.015$ and $x = 0.015Q$ ceramics. In the whole test temperature range, all samples exhibited saturated P - E loops and sharp I - E peaks. As the test temperature increased, the observed P_r and P_{max} increased, while the E_C decreased, accompanied by the thinning of the P - E loops and a sharper I - E peak. The main reason is that the domain wall pinning effect is significantly weakened by the thermal activation energy. At lower temperatures, defects within the material strongly pin the domain walls, hindering their movement, resulting in a higher required electric field strength for polarisation reversal and some electric domains not being able to fully reverse, which causes a lower P_r . As the temperature rises, the thermal disturbance effect energises the domain wall movement, which makes it easier to overcome the pinning barrier of defects, so that the domain wall can start and complete a wide

range of collaborative turnover at a lower electric field, which directly leads to the reduction of E_C . Simultaneously, more released electric domains can align fully along the electric field direction, enhancing polarisation efficiency and increasing P_r . This results in a finer and narrower P - E loop, and in obtaining I - E curves with higher current peaks. And combined with the fitting results of XPS, it can be seen that the defect content (such as $V_O^{\bullet\bullet}$) in the quenched samples has significantly decreased, resulting in a weakened pinning effect of domain walls. As the test temperature increases, which results in higher P_r and lower E_C .

To real-time characterise the dynamic variation of the piezoelectric coefficient of BF-BT ceramics under a temperature field, the in-situ d_{33} testing technology was adopted in this study. This technology can effectively reveal the evolution characteristics and thermal stability of the piezoelectric response of BF-BT ceramics with temperature, which is of great guiding significance for evaluating the practical potential and application feasibility of the materials. Fig. 8(a) presents the in-situ depolarisation curve of the 0.72BF-0.28BT-0.015BST-xBZT ($x = 0.005$, 0.015, 0.015Q) ceramics, showing the real-time piezoelectric properties of ceramics at high temperature. The piezoelectric properties of 0.72BF-0.28BT-0.015BST-0.015BZT ceramics were significantly improved after quenching treatment, in which the room-temperature piezoelectric constant (d_{33}) has increased from 182 pC/N to 196 pC/N. Additionally, the in-situ depolarisation test results show that the quenching treatment has also effectively improved the high-temperature piezoelectric properties and thermal stability of the ceramics, the *In-situ* d_{33} increasing from 314 pC/N to 378 pC/N, and the T_{dr} was enhanced from 284 °C to 308 °C. The piezoelectric coefficient is expressed as follows:

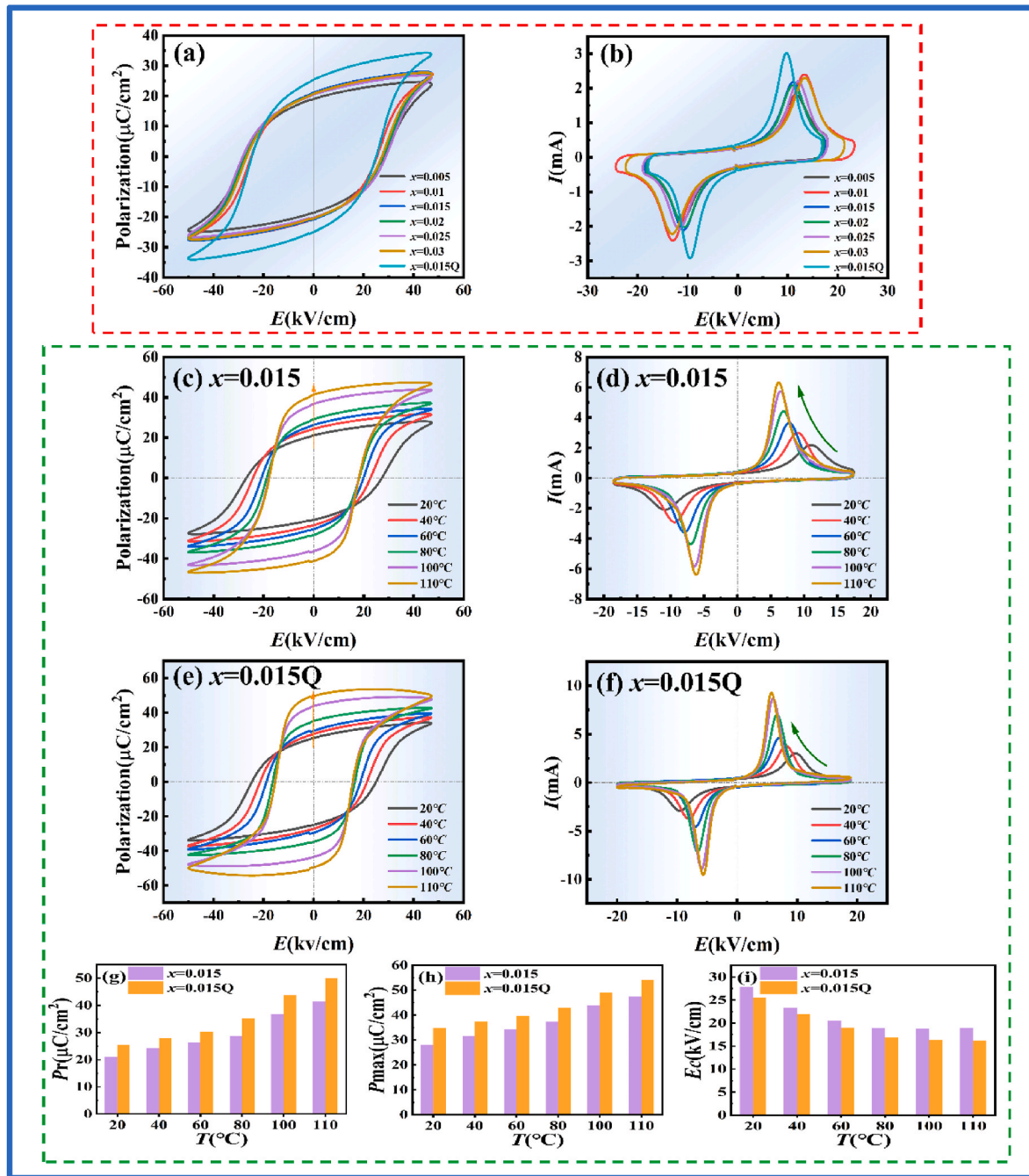


Fig. 7. (a) P - E hysteresis loop and (b) I - E current curve of 0.72BF-0.28BT-0.015BST- x BZT ($0.005 \leq x \leq 0.03$, 0.015Q) ceramic at 50 kV/cm electric field. The P - E curves and I - E curves of (c,d) 0.72BF-0.28BT-0.015BST-0.015BZT ceramics and (e,f) 0.72BF-0.28BT-0.015BST-0.015BZT ceramics after quenching under the test conditions of electric field 50 kV/cm, frequency 1Hz and temperature range 20 °C-110 °C. (g-i) The temperature-dependent diagrams of P_r , P_{max} and E_c for the $x = 0.015$ and $x = 0.015Q$ ceramics.

$$d_{33} = 2Q\epsilon P_s \quad (3)$$

Where Q represents the electrostrictive coefficient, ϵ is the dielectric constant, and P_s represents the spontaneous polarisation intensity. The analysis of the dielectric temperature spectra shows that the quenching process improves the dielectric constant (ϵ) of ceramics in the high-temperature region, which leads to the enhancement of high-temperature piezoelectric activity.

Fig. 8(b) shows the evolution of the electric domains in the 0.72BF-0.28BT-0.015BST-0.015BZT quenched sample during the depolarisation process. In the in-situ depolarisation curve of the ceramic sample exists a stable stage and two ascending stages. The piezoelectric effect of

perovskite-type ferroelectric ceramics stems from the combined contribution of both intrinsic and extrinsic mechanisms [37]. The first rise period of piezoelectric activity occurs between room temperature and 70 °C, and the introduction of different heterovalent cations leads to lattice deformation, which is essential to enhance the intrinsic contribution of the ceramics. This further triggers the soft ferroelectric effect, which shows that the coercive field (E_c) is reduced under the applied electric field, making the ferroelectric domain reversal easier, and finally, the piezoelectric coefficient is greatly increased [38]. Within the temperature range of 70-160 °C, there exists a stable performance interval, which may be due to the dynamic competition between the decomposition of nano-domains and PNRs. This competitive mechanism

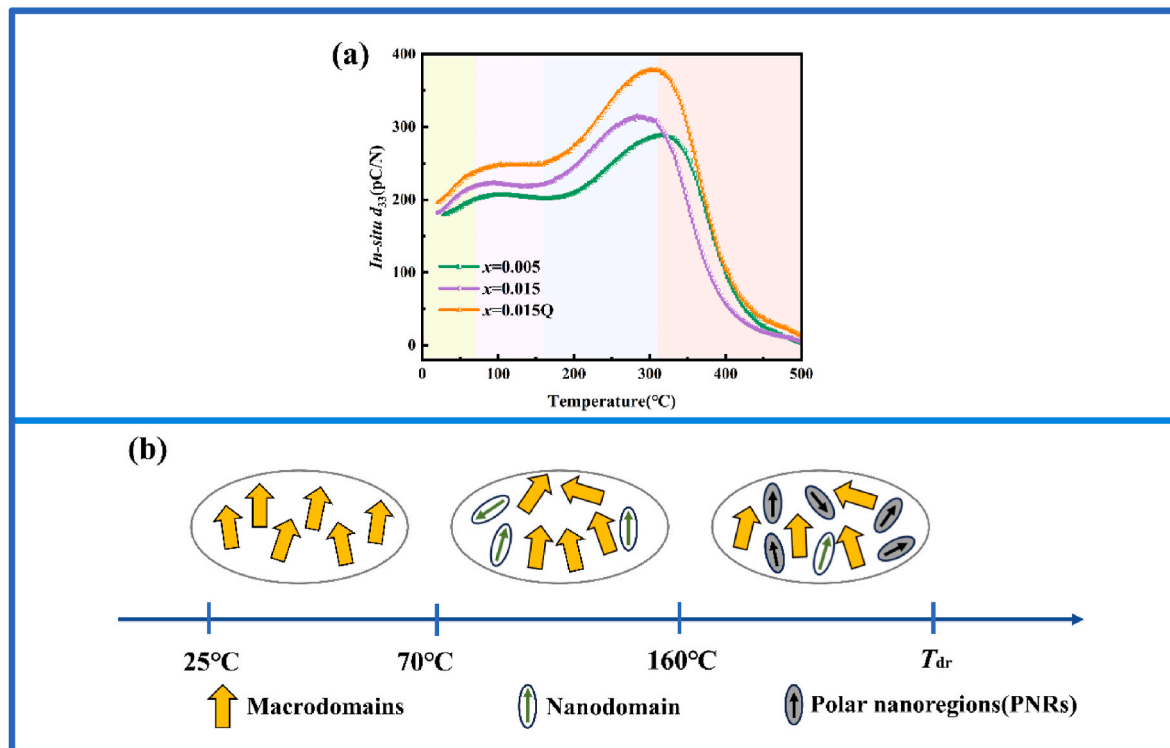


Fig. 8. (a) In-situ depolarisation curve of 0.72BF-0.28BT-0.015BST- x BZT ($x = 0.005, 0.015, 0.015Q$) ceramics; (b) the evolution diagram of the electric domains during the depolarisation process of the 0.72BF-0.28BT-0.015BST-0.015BZT quenched sample.

ensures the stability of the piezoelectric coefficient of the material within this temperature range [39]. The second piezoelectric activity enhancement stage occurs at $160\text{ }^{\circ}\text{C}$ - T_{dr} , which is attributed to the gradual evolution of the nanodomain structure into PNRs, where the process reduces the polarisation inversion barrier within the system, accelerating the dynamic response of the micro-region and ultimately leading to a significant enhancement in the piezoelectric performance.

Transmission electron microscopy (TEM) is one of the core characterisation methods to reveal the nature of piezoelectric properties of BF-BT ceramics and analyse their regulation mechanisms. By accurately capturing the intrinsic microstructure of materials, it directly correlates and determines the quality of piezoelectric properties, providing irreplaceable microcosmic support for the directional regulation, optimisation, and mechanism explanation of piezoelectric properties. It forms a "micro-macro" complementarity with the macro piezoelectric response characterised by the in-situ d_{33} test, jointly improving the research system of piezoelectric properties of BF-BT ceramics. Fig. 9 shows the test results of 0.72BF-0.28BT-0.015BST-0.015BZT transmission electron diffraction (TEM). Fig. 9(a) shows the magnified image of the lattice fringes of the test sample and the corresponding selected area electron diffraction (SAED) pattern. It can be seen from the figure that the lattice spacing of the sample is $d = 0.231\text{ nm}$. Fig. 9(b) observes the obvious local nano-domain structure. Fig. 9(c) shows the clear Moiré fringes (orange dashed box), whose magnetic region structure can be attributed to polar nanoregions (PNRs), and their formation is likely due to the enhanced disorder in composition caused by ion doping substitution. The existence of PNRs can reduce the energy barrier of domain reversal, which promotes the deflection response of the domain under an electric field. At the same time, the structure also enhances the response of the material to the external temperature field, which is beneficial for maintaining high piezoelectric performance of ceramics in high-temperature environments [11]. Fig. 9(d) represents some nano-domain structures, which play a dominant role in achieving a large piezoelectric response and maintaining temperature stability.

4. Conclusion

In summary, the 0.72BF-0.28BT-0.015BST- x BZT ($0.005 \leq x \leq 0.03$) ceramics were successfully prepared by the traditional solid-state sintering method. Through the multi-component optimisation strategy, the best performance ($d_{33} = 182\text{ pC/N}$, $T_C = 529\text{ }^{\circ}\text{C}$, $\text{In-situ } d_{33} = 314\text{ pC/N}$, $T_{dr} = 284\text{ }^{\circ}\text{C}$) was obtained at $x = 0.015$. The XRD results indicate that the addition of BZT results in the morphotropic phase boundaries structure for the 0.72BF-0.28BT-0.015BST ceramic. The TEM results show that the ceramic possesses a local heterogeneous structure where both nanodomains and polar nanoregions coexist. Quenching treatment can more effectively inhibit the formation of defects, enhance the long-range domain order, and promote the transformation of ceramic samples to normal ferroelectric. These results suggest that the process improves the high-temperature piezoelectricity and thermal stability of ceramics, and the quenched samples with $x = 0.015Q$ show better comprehensive properties ($d_{33} = 196\text{ pC/N}$, $T_C = 540\text{ }^{\circ}\text{C}$, $\text{In-situ } d_{33} = 378\text{ pC/N}$, $T_{dr} = 308\text{ }^{\circ}\text{C}$). These findings provide a new way to prepare lead-free piezoelectric ceramics with both high piezoelectric activity and high Curie temperature.

CRediT authorship contribution statement

Xiaohui Pan: Writing – review & editing, Writing – original draft. **Xinyi Yang:** Formal analysis. **Haitao Zhang:** Data curation. **Xinyi Zhou:** Data curation. **Hua Tan:** Methodology. **Huabin Yang:** Methodology, Funding acquisition. **Minhong Jiang:** Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

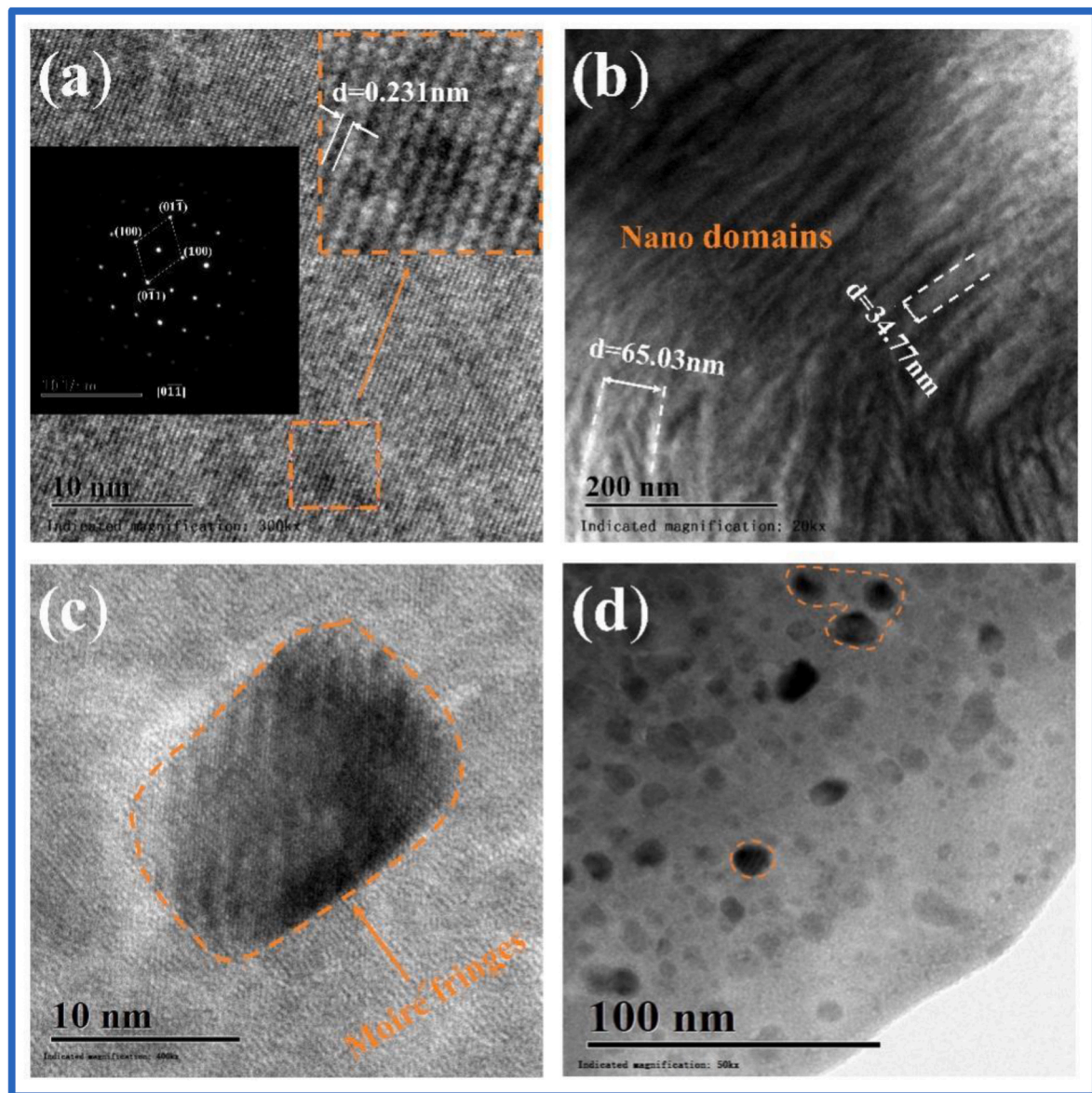


Fig. 9. TEM images of the 0.72BF-0.28BT-0.015BST-0.015BZT ceramic with nanodomains.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grants 52162016, 52202133), Guangdong Basic and Applied Basic Research Foundation (2023A1515140060, 2024A1515140014), Guangdong HUST Industrial Technology Research Institute and Guangdong Provincial Key Laboratory of Manufacturing Equipment Digitization (2023B1212060012).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mssp.2026.110559>.

Data availability

Data will be made available on request.

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